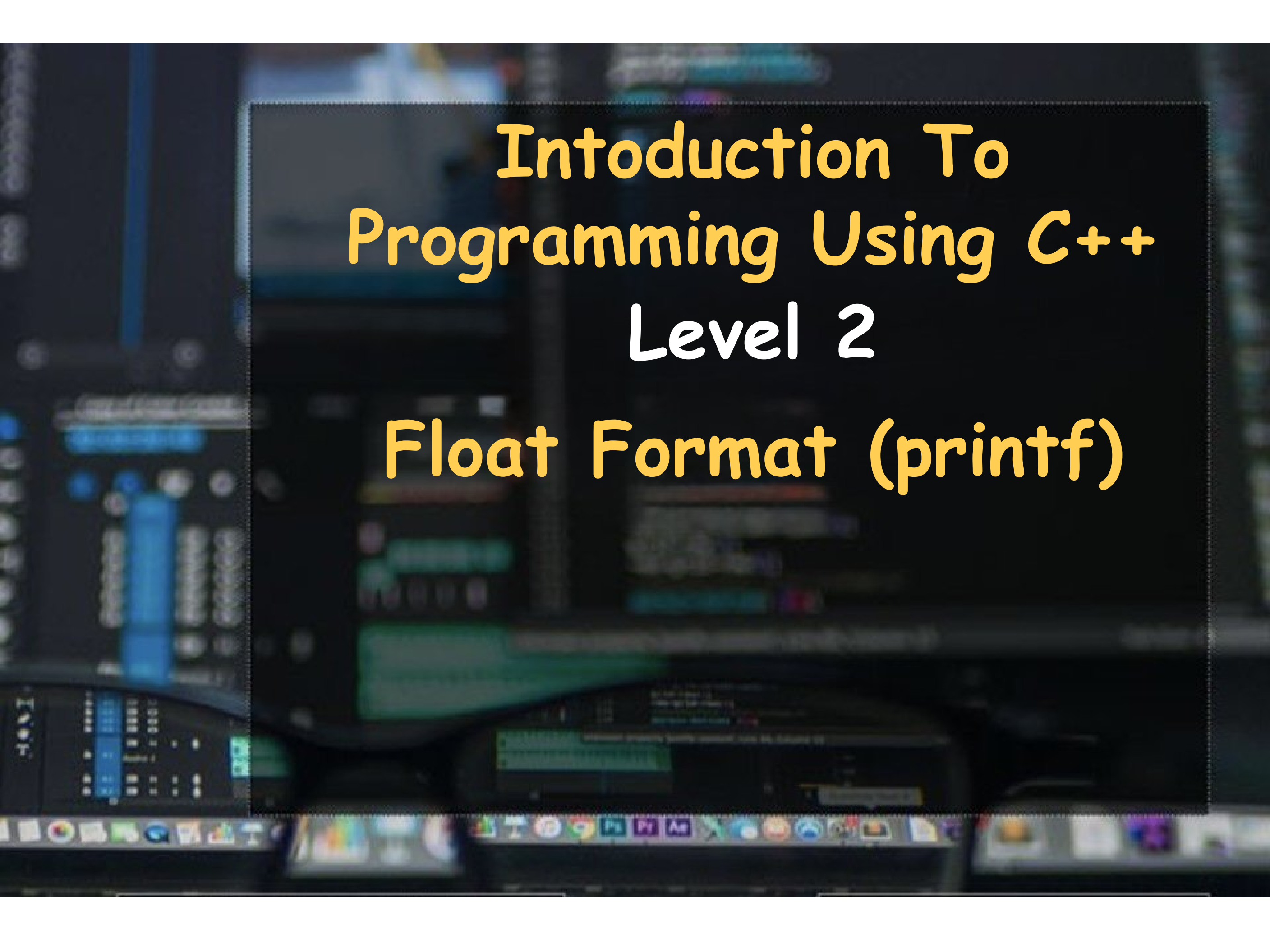
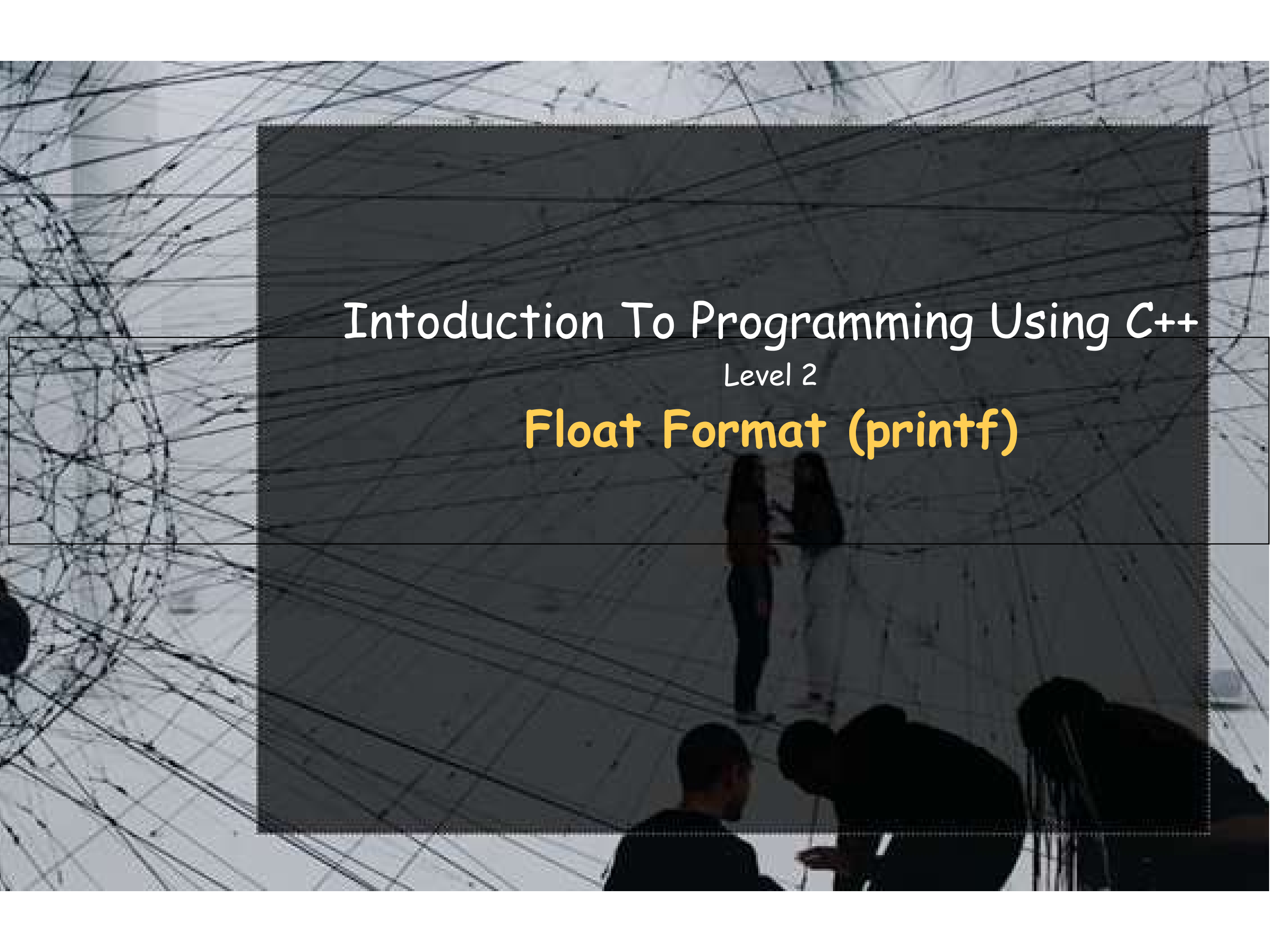


The background of the slide is a blurred image of a computer monitor. On the left side, a calculator application is visible, showing a numeric keypad and some text. The main area of the monitor displays lines of code in a dark-themed editor, with some text highlighted in green. At the bottom of the screen, a macOS-style dock is visible with various application icons including Finder, Safari, and others.

Introduction To Programming Using C++ Level 2

Float Format (printf)



Introduction To Programming Using C++

Level 2

Float Format (printf)

Definition

- ◆ The ``%f`` Format Specifier is used with ``printf()`` to print Floating-Point Numbers.
- ◆ You can control the number of digits displayed after the Decimal Point using a Precision Specification.
- ◆ The Precision can be provided directly or dynamically through another argument.

Explanation

 There are two useful forms of Float Formatting:

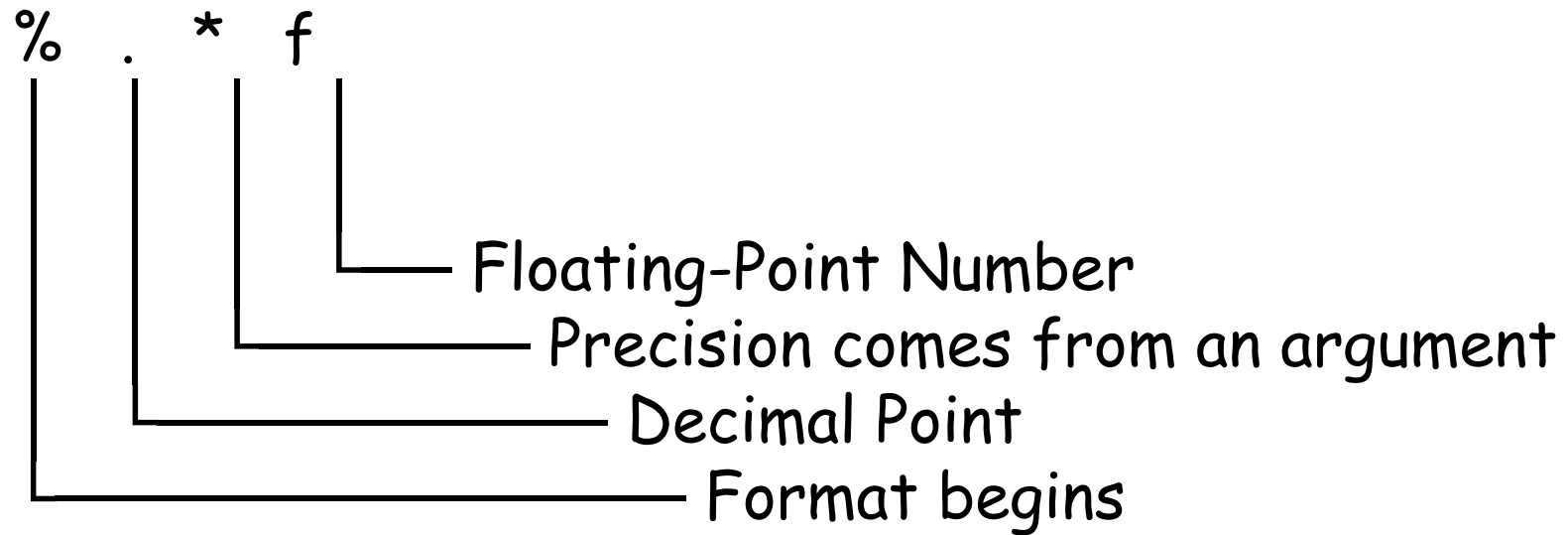
 ``%.3f``

✓ Prints exactly 3 digits after the Decimal Point.

 ``%.*f``

✓ The number of digits after the Decimal Point is provided through another argument.

Procedures



💡 Example

```
using namespace std;
int main()
{
    float PI = 3.14159265;
    // Precision specification
    printf("Precision specification of %.*f\n", 1, PI);
    double d = 12.45;
    // Fixed precision
    printf("The double value is : %.3f\n", d);
    return 0;
}
```

➡ Result

Precision specification of 3.1
The double value is : 12.450

Note

- ◆ The Precision specifies the number of digits displayed after the Decimal Point.
- ◆ If the original value has fewer digits than the requested Precision, `printf()` adds zeros.

⊘ Warning

- ◆ Do not confuse Precision with the total number of digits in the number.
- ◆ In ``%.3f``, the ``3`` means 3 digits after the Decimal Point, not 3 digits in total.



Comparison

▨ ``%.3f``

- ✓ Uses a fixed Precision.
- ✓ Always displays 3 digits after the Decimal Point.
- ✓ Example: ``12.45` → `12.450``

▤ ``%.*f``

- ✓ Uses a dynamic Precision.
- ✓ The Precision is provided as an additional argument.
- ✓ Example: ``printf("%.*f", 1, PI)` → `3.1``

✦ Summary

- ✓ ``%f`` is used to print Floating-Point Numbers with ``printf()``.
- ✓ ``%.3f`` prints exactly 3 digits after the Decimal Point.
- ✓ ``%.*f`` allows the Precision to be provided dynamically.
- ✓ The ``*`` means that the Precision comes from an additional argument.
- ✓ Precision controls the number of digits displayed after the Decimal Point.
- ✓ ``printf()`` can add trailing zeros when necessary.

Thank you 😊